Pandas Coding Assignment (Answers)

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```
import pandas as pd
```

```
df=pd.read_csv("/content/Salaries.csv")
```

```
/tmp/ipykernel_3236/2769251863.py:1: DtypeWarning: Columns (3,4,5,6,12) have mixed types. Specify d
  df=pd.read_csv("/content/Salaries.csv")
```

```
#Q3
df.head()
```

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	TotalPay	TotalPayBenefits
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	0.0	400184.25	NaN	567595.43	567595.43
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02	245131.88	137811.38	NaN	538909.28	538909.28
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.6	NaN	335279.91	335279.91
3	4	CHRISTOPHER CHONG	WIRE ROPE CABLE MAINTENANCE MECHANIC	77916.0	56120.71	198306.9	NaN	332343.61	332343.61
4	5	PATRICK GARDNER	DEPUTY CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	134401.6	9737.0	182234.59	NaN	326373.19	326373.19

```
#Q4
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148654 entries, 0 to 148653
Data columns (total 13 columns):
 #   Column            Non-Null Count   Dtype  
---  ------            --------------   -----  
 0   Id                148654 non-null  int64  
 1   EmployeeName      148654 non-null  object 
 2   JobTitle          148654 non-null  object 
 3   BasePay           148049 non-null  object 
 4   OvertimePay       148654 non-null  object 
 5   OtherPay          148654 non-null  object 
 6   Benefits          112495 non-null  object 
 7   TotalPay          148654 non-null  float64
 8   TotalPayBenefits  148654 non-null  float64
 9   Year              148654 non-null  int64  
 10  Notes             0 non-null       float64
 11  Agency            148654 non-null  object 
 12  Status            38119 non-null   object 
dtypes: float64(3), int64(2), object(8)
memory usage: 14.7+ MB
```

```
#Q5
df["BasePay"] = pd.to_numeric(df["BasePay"], errors='coerce')
average_base_pay = df["BasePay"].mean()
print(average_base_pay)
```

```
66325.4488404877
```

```
#Q6
df["OvertimePay"]=pd.to_numeric(df["BasePay"], errors='coerce')
highest_overtime_pay=df["OvertimePay"].max()
print(highest_overtime_pay)
```

```
319275.01
```

```
print(df.columns)
```

```
Index(['Id', 'EmployeeName', 'JobTitle', 'BasePay', 'OvertimePay', 'OtherPay',
       'Benefits', 'TotalPay', 'TotalPayBenefits', 'Year', 'Notes', 'Agency',
       'Status'],
      dtype='object')
```

```
print(df.loc[df["EmployeeName"] == "JOSEPH DRISCOLL", "JobTitle"])
```

```
24    CAPTAIN, FIRE SUPPRESSION
Name: JobTitle, dtype: object
```

```
#Q8
total_pay_benefits = df[df["EmployeeName"] == "JOSEPH DRISCOLL"]["TotalPayBenefits"]
print(total_pay_benefits)
```

```
24    270324.91
Name: TotalPayBenefits, dtype: float64
```

```
#Q9
highest_paid_df = df.sort_values(by="TotalPayBenefits", ascending=False).head(1)
display(highest_paid_df)
```

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	TotalPay	TotalPayBenefits
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	167411.18	400184.25	NaN	567595.43	567595.43

```
#Q10
df.sort_values(by="TotalPayBenefits", ascending=True).head(1)
```

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	TotalPay	TotalPayBenefits
148653	148654	Joe Lopez	Counselor, Log Cabin Ranch	0.0	0.0	-618.13	0.00	-618.13	-618.13

Yes, the strange thing is that the person's total pay benefits is a negative value (-618.13). Normally, earnings are positive or zero, so having a negative total pay is quite unusual.

```
#Q11
df.groupby("Year")["TotalPay"].mean()
```

Year	TotalPay
2011	71744.103871
2012	74113.262265
2013	77611.443142
2014	75463.918140

dtype: float64

```
#Q12
df["JobTitle"].nunique()
```

```
2159
```

```
#Q13
df["JobTitle"].value_counts().head(5)
```

JobTitle	count
Transit Operator	7036
Special Nurse	4389
Registered Nurse	3736
Public Svc Aide-Public Works	2518
Police Officer 3	2421

dtype: int64

```
#Q14
df[df["JobTitle"].str.contains("Chief", case=False, na=False)].shape[0]
```

```
627
```

```
#Q15
```

```
df["Len_JobTitle"] = df["JobTitle"].apply(lambda x: len(str(x)))
df[["Len_JobTitle", "TotalPayBenefits"]].corr()
```

	Len_JobTitle	TotalPayBenefits
Len_JobTitle	1.000000	-0.036878
TotalPayBenefits	-0.036878	1.000000

```
df.columns
```

```
Index(['Id', 'EmployeeName', 'JobTitle', 'BasePay', 'OvertimePay', 'OtherPay',
       'Benefits', 'TotalPay', 'TotalPayBenefits', 'Year', 'Notes', 'Agency',
       'Status', 'title_len', 'Len_JobTitle'],
      dtype='object')
```

```
#Answer: There is almost no correlation between the length of the job title and the salary (TotalP
```

Напишите программный код или сгенерируйте его с помощью искусственного интеллекта.

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